MIL-P-46620C(MU)
28 January 1977
SUPERSEDING
MIL-P-46620B
28 May 1970

# MILITARY SPECIFICATION

## PROJECTILE, 20MM, HEI, M97A2; METAL PARTS

This specification is approved for use by all Departments and Agencies of the Department of Defense.

## 1. SCOPE

1.1 Scope. This specification covers the metal parts for 20mm high explosive incendiary projectiles, M97A2. (see 6.1)

## 2. APPLICABLE DOCUMENTS

2.1 Issue of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

| | |
|---|---|
| TT-C-490 | - Cleaning Methods and Pretreatment of Ferrous Surfaces for Organic Coatings |

MILITARY

| | |
|---|---|
| MIL-A-2550 | - Ammunition, General Specification for |

STANDARDS

MILITARY

| | |
|---|---|
| MIL-STD-105 | - Sampling Procedures and Tables for Inspection by Attributes |
| MIL-STD-109 | - Quality Assurance Terms and Definitions |
| MIL-STD-651 | - Visual Inspection Standards for 20mm Ammunition and Components |

FSC 1305

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Frankford Arsenal, ATTN: SARFA-MDM, Philadelphia, Pennsylvania 19137 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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2.1 (Cont)

MIL-STD-1168 - Ammunition Lot Numbering
MIL-STD-1169 - Packaging, Packing and Marking for Shipment of Inert Ammunition Components
MIL-STD-1235 - Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes

DRAWINGS

ARMAMENT COMMAND

F7258861 - Carton, for Interplant Shipment of Projectiles, 20mm, Assemblies and Metal Parts Assemblies (200)
D7259066 - Projectile, 20mm, HEI, M97A2, Metal Parts
LI 7259066 - Index of Inspection Equipment Lists for: Projectile, 20mm, HEI, M97A2, Metal Parts
D7259090 - Cartridge, 20mm, for Projectile Metal Parts Test
D7259136 - Shipping Container for Inert Projectile, 20mm, Assemblies and Metal Parts Assemblies (10,000)

PUBLICATIONS

U.S. ARMY MATERIEL COMMAND

AMCR 715-505 - Ammunition Ballistic Acceptance Test Methods, Volume 8, Test Procedures for 20mm Cartridges

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

American Society for Testing and Materials (ASTM)
ASTM B117-73 - Method of Test for Salt Spray (Fog) Testing

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103)

Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.



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## 3. REQUIREMENTS

3.1 General. The parts and assemblies shall be in accordance with all the requirements of Drawing (Dwg.) D7259066 and with all requirements of the drawings and specifications shown thereon.

3.2 Manufacturing process. All parts and assemblies shall be manufactured by a process approved by the contracting officer and no deviation from that process shall be made without his prior approval. (see 6.5)

3.3 First article sample. Unless otherwise directed by the contracting officer, a first article sample shall be required.

3.4 Banding. The rotating band blank and the rotating band seat shall be clean at the time of banding. Shot or sandblasting shall not be used to clean the rotating band seat after knurling. The rotating band blank shall fill the band seat.

3.5 Protective coating.

3.5.1 Phosphate coating. The pretreatment process used shall be in accordance with the applicable requirements of Dwg. D7259066. The phosphate coating shall be in accordance with the requirements of TT-C-490 for appearance and minimum weight, as applicable to the type of phosphate coating.

3.5.2 Paint coating. The interior and exterior coatings of the projectile metal parts, shall be in accordance with the applicable requirements of Dwg. D7259066. The exterior paint adhesion shall be in accordance with the applicable requirements of TT-C-490.

3.5.3 Corrosion resistance. The exterior painted surface of the projectile metal parts, shall show no evidence of corrosion (red rust) after exposure to a 5% solution of atomized salt spray for 24 hours.

3.6 Metal parts security. The sample projectile metal parts, inert charged and assembled into cartridges in accordance with the requirements of Dwg. D7259090 shall not separate or break up in the test for metal parts security.

3.7 Workmanship. The requirements for workmanship are as specified by applicable drawings, referenced specifications and the following:

3.7.1 Processing defects. The projectile metal parts shall be free of defects such as dents, scratches, cuts, sharp edges, burrs, and other forms of substandard workmanship.

3.7.2 Cleanliness. The projectile metal parts shall be free of all foreign matter.



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## 4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Quality assurance terms and definitions. Reference shall be made to MIL-STD-109 to define quality assurance terms.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

1. First article inspection (see 4.3).
2. Quality conformance inspection (see 4.4).

4.3 First article inspection.

4.3.1 First article sample. The sample shall be delivered for inspection in accordance with first article approval provisions of the contract and MIL-A-2550, and shall consist of:

a. Inspection and nonfiring tests:

| | | |
|---|---|---|
| (1) | 3 | Standard test panels |
| (2) | 32 | Rotating bands |
| (3) | 32 | Base covers |
| (4) | 125 | Bodies (prior to banding) |
| (5) | 125 | Body, rotating band assemblies (uncoated) |
| (6) | 125 | Projectile, metal parts (uncoated) |
| (7) | 125 | Projectile, metal parts (coated prior to paint) |
| (8) | 125 | Projectile, metal parts (painted) |

b. Firing test:

4,000 Projectile, metal parts

The projectile metal parts shall be identified in accordance with MIL-STD-1168.

4.3.1.1 Inspection. After inspection by the contractor and provisional acceptance by the Government activity, the sample will be inspected in accordance with the provisions of MIL-A-2550. Failure of the original first article sample to be in accordance with the applicable drawings and the requirements of Para. 3, as specified herein, will result in sample disapproval. (see 6.6)



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4.4 Quality conformance inspection.

4.4.1 Submission of product. The product shall be submitted in accordance with MIL-STD-105 or MIL-STD-1235, as applicable.

4.4.1.1 Lot. A lot shall consist of projectile metal parts produced by one manufacturer in one unchanged process, in accordance with the contract and the same drawings and drawing revisions, and same specification and specification revision.

4.4.1.2 Lot identification. Each projectile metal parts and each packed lot shall be identified in accordance with applicable drawings and MIL-STD-1168.

4.4.2 Examination. One hundred percent examination shall be performed by the contractor for all critical defects. Examination for major and minor defects shall be performed as specified herein. Each part or assembly found to be defective shall be rejected.

4.4.2.1 Sampling and acceptance criteria. The sampling plans and procedures shall be in accordance with MIL-STD-105 or MIL-STD-1235. To determine product acceptability, major or minor defects as listed herein may be considered collectively on a class basis or they may be considered individually. However, where major defects are considered collectively on a class basis, the acceptance number for any individual defect of the class shall be limited by the assigned Acceptable Quality Level (AQL) for individual defect associated with the class. (see 4.4.2.2.c) In addition, where three or less defects appear in a classification, or, where MIL-STD-1235 is used, sampling plans applicable to a class shall not be used for major defects. (see 6.3)

4.4.2.2 AQL's. The following AQL's are assigned to major and minor defects:

a. Individual:

| | |
|---|---|
| Major | 0.25% |
| Minor | 0.40% |

b. Class basis:

| | |
|---|---|
| Major | 1.50% |
| Minor | 2.50% |

c. Individual defects associated with the class basis:

| | |
|---|---|
| Major | 0.40% |

4.4.3 Classification of defects 1/. The classification of defects shall be as specified herein: (see 6.4)



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4.4.3.1 Rotating band blank (see Dwg. B7258816, a detail of Dwg. D7259066).

| Categories and Defects | Method of Inspection |
|---|---|
| Critical: None defined. | |
| Major: | |
| 101. Foreign matter | Visual (see 3.4) |
| 102. Defective metal (porosity, lamination or inclusion) | Visual |
| 103. Incomplete or deformed | Visual 2/ |
| Minor: | |
| 201. Width, max. | Gage |

4.4.3.2 Cover, base (see Dwg. B7259155, a detail of Dwg. D7259066).

| Categories and Defects | Method of Inspection |
|---|---|
| Critical: None defined. | |
| Major: | |
| 101. Diameter | Gage |
| Minor: | |
| 201. Thickness, min. | Gage |
| 202. Thickness, max., including out of flatness | Gage |
| 203. Foreign matter | Visual (see 3.7.2) |

4.4.3.3 Body and rotating band assembly (prior to banding) (see Dwg. D7259067, a detail of Dwg. D7259066).

| Categories and Defects | Method of Inspection |
|---|---|
| Critical: None defined. | |
| Major: | |
| 101. Length, mouth to rear edge of rotating band seat | Gage |
| 102. Knurling missing or incomplete | Visual |
| 103. Depth, knurling, min. | Visual/gage 3/ |
| 104. Foreign matter in rotating band seat | Visual (see 3.4) |
| Minor: | |
| 201. Diameter, rotating band seat (knurled) | Gage |
| 202. Width, rotating band seat | Gage |
| 203. Width, knurling, min. | Visual/Gage 3/ |

4.4.3.4 Body and rotating band assembly (after banding) (see Dwg. D7259067, a detail of Dwg. D7259066).

| Categories and Defects | Method of Inspection |
|---|---|
| Critical: None defined. | |



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4.4.3.4 (Cont)

| Categories and Defects | Method of Inspection |
|---|---|
| **Major:** | |
| 101. Thickness, base, at drill point (if present), min. | Gage |
| 102. Diameter, base cover recess, max. | Gage |
| 103. Height, base cover crimping lip, min. | Gage |
| 104. Position, base cover recess, with datum surface | Gage |
| 105. Length, mouth to base cover seat, max. | Gage |
| 106. Thickness, base (except drill point), min. | Gage |
| **Minor:** | |
| 201. Diameter, base cover recess, min. | Gage |
| 202. Height, base cover crimping lip, max. | Gage |
| 203. Foreign matter in base cover recess | Visual |
| 204. Length, mouth to base cover seat, min. | Gage |
| 205. Width, crimping lip flat | Gage |
| 206. Angle of crimping lip | Gage |

4.4.3.5 Projectile, metal parts (uncoated) (see Dwg. D7259066 and D7259067, a detail of Dwg. D7259066).

| Categories and Defects | Method of Inspection |
|---|---|
| **Critical:** | |
| 1. Crack, seam or fold in body | Visual |
| 2. Base cover missing, incomplete, multiple, loose, or incorrectly assembled and crimped | Visual/Manual |
| **Major:** | |
| 101. Diameter, bourrelet, min. | Gage |
| 102. Diameter, body, rear of rotating band, min. | Gage |
| 103. Runout, body, rear of rotating band, with datum surface | Gage |
| 104. Width, rotating band | Gage |
| 105. Diameter, rotating band | Gage |
| 106. Length, mouth to rear edge of rotating band | Gage |
| 107. Runout, rotating band with datum surface | Gage |
| 108. Void at junction of rotating band and projectile body | Visual |
| 109. Threads not full | Visual |
| 110. Diameter, body cavity | Gage |
| 111. Runout, body cavity, with datum surface | Gage |
| 112. Profile, crimp groove | Gage |
| 113. Radius, rear edge of crimp groove, max. | Gage |
| 114. Location, crimp groove | Gage |
| 115. Mutilated | Visual |
| 116. Corrosion | Visual |



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4.4.3.5 (Cont)

| Categories and Defects | Method of Inspection |
|---|---|
| **Minor:** | |
| 201. Depth, thread counterbore, if present, max. | Visual/Gage 4/ |
| 202. Diameter, thread counterbore, if present, max. | Visual/Gage 4/ |
| 203. Outside diameter, mouth end | Gage |
| 204. Finish improper | Visual |
| 205. Depth and width of undercut, front and rear of rotating band | Gage |
| 206. Break, mouth, at ogive | Visual/Gage 4/ |

4.4.3.6 Projectile, metal parts (coated, prior to painting) (see Dwg. D7259066 and D7259067, a detail of Dwg. D7259066).

| Categories and Defects | Method of Inspection |
|---|---|
| **Critical:** None defined. | |
| **Major:** | |
| 101. Pitch diameter, thread | Gage |
| 102. Runout, thread with datum surface | Gage |
| **Minor:** | |
| 201. Minor diameter, thread | Gage |
| 202. Length, complete thread, min. | Gage |

4.4.3.7 Projectile, metal parts (painted) (see Dwg. D7259066).

| Categories and Defects | Method of Inspection |
|---|---|
| **Critical:** None defined. | |
| **Major:** | |
| 101. Length, overall, max. | Gage |
| 102. Diameter, bourrelet, max. | Gage |
| 103. Diameter, body, rear of rotating band, max. | Gage |
| 104. Mutilated | Visual |
| 105. Foreign matter in cavity or threads | Visual |
| 106. Weight | Balance |
| **Minor:** | |
| 201. Interior or exterior paint inadequate | Visual |
| 202. Exterior paint not dry | Manual |
| 203. Paint on rotating band | Visual |
| 204. Paint on threads | Visual |
| 205. Marking incorrect, incomplete, illegible or missing | Visual |
| 206. Rotating band damaged | Visual |
| 207. Foreign matter on exterior | Visual |

1/ MIL-STD-651 shall apply in defining and evaluating visual defects.



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2/ A rotating band blank which is deformed or is judged to lack sufficient metal in any part of its width or thickness to provide for complete band seat fill and minimum rotating band profile dimensions shall be classed defective.

3/ Measurement of knurling for compliance with the minimum depth and width requirements shall apply when visual inspection indicates that either dimension is borderline.

4/ Measurement of the diameter and depth of the thread counterbore or measurement of mouth break at ogive shall apply where visual inspection indicates that the dimension is borderline.

4.4.4 Tests. The tests listed in Table I shall be performed on each projectile lot in accordance with the test methods prescribed in 4.5. Only projectile metal parts having met the visual and dimensional requirements and having been selected in such a manner that the sample is representative of the entire lot shall be used in the tests. The selected projectile metal parts shall be thoroughly mixed before being divided into samples for the various tests. Sample size and acceptance criteria for each test shall be as specified.

Table I

| Test | Sample Size | Item | Rqmt Para |
|---|---|---|---|
| Band tightness | 1/ 315 | Projectile, metal parts, painted or uncoated | 3.4 |
| Phosphate coating (Appearance and weight) | 2/ -- | 3 standard panels (see TT-C-490) | 3.5.1 |
| Paint adhesion | 3/ 13 | Projectile, metal parts, painted | 3.5.2 |
| Corrosion resistance | 4/ 32 | Projectile, metal parts, painted | 3.5.3 |
| Metal parts security | 5/ 250 | Projectile, metal parts, painted | 3.6 |

1/ If two or more units of the sample have an average radial void, between the seated band and the band seat, greater than 0.004 inch, the lot shall be rejected. If one unit of the sample has an average radial void greater than 0.004 inch, a second sample consisting of the same number of units as specified for the first sample shall be tested. If, in the accumulated samples, two or more units have an average radial void greater than 0.004 inch, the lot shall be rejected.

2/ Failure of the phosphate coating on the standard panels to comply with the applicable requirements, shall cause rejection of all projectiles phosphated since the preceding test.

3/ If two or more units of the sample fail to comply with the requirement for paint adhesion, the lot shall be rejected. If one unit of the sample fails to comply with the requirement, a second sample consisting of the same number of units as specified for the first sample shall be tested. If in the accumulated samples, two or more units fail to comply with the requirement for paint adhesion, the lot shall be rejected.



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4/ If two or more units of the sample fail to comply with the corrosion resistance requirement, the lot shall be rejected. If one unit fails to comply with the requirement, a second sample consisting of the same number of units as specified for the first sample shall be tested. If in the accumulated samples, two or more units fail to comply with the corrosion resistance requirement, the lot shall be rejected.

5/ The occurrence of one or more of the following casualties, as witnessed by hole(s) in the fragmentation screen(s), or recovery of the part(s) or fragment(s), shall be cause for rejection of the lot, except as otherwise provided:

a. Body breakup.

b. Separation of the base cover from the projectile body.

c. Complete rotating band separation. This is defined as detachment of all or part of the band having band seat knurling imprint. The following acceptance criteria shall apply to detached band fragments having no visible evidence of band seat knurling imprint:

(1) The lot shall be accepted if two or less projectiles throw band fragments having length not greater than 0.500 inch.

(2) The lot shall be rejected if three or more assemblies throw band fragments having length not greater than 0.500 inch.

(3) The lot shall be rejected if any projectile throws a band fragment having length greater than 0.500 inch (indicative of deep seated band lamination).

(4) The lot shall not be penalized for band particles thrown as a result of normal band fringing or slivering.

4.4.5 Packaging and marking inspection. Inspection for packaging, packing and marking shall be in accordance with applicable requirements of MIL-STD-1169.

4.4.6 Inspection equipment. Index of Inspection Equipment List (LI) 7259066 identifies the applicable inspection equipment required to perform examination and tests as specified herein. Equipment design(s) shall be in accordance with the applicable code designations. The code designations are defined on Dwg. B11075228, a detail of LI 7259066. The provisions of MIL-A-2550 shall apply.

4.5 Test methods and procedures.



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4.5.1 Band tightness. The method of test shall comply with all requirements of the operating instructions for the band tightness tester (Dwg. B7259545, a detail of LI 7259066). The inspection sample shall be cycled through the tester to determine compliance with the rotating band tightness requirement. The total measurement of radial void, registered on the tester indicator at final ram pressure, shall be divided by four (number of indentors) to determine the average radial void.

4.5.2 Phosphate coating. The method of test shall be as prescribed by TT-C-490 for the type of coating used (see Dwg. D7259066). The prescribed panels shall be processed, with the assemblies to be represented, at the start of daily production and at 4-hour intervals, maximum, thereafter. Appearance and weight of the coating shall be recorded.

4.5.3 Paint adhesion. The method of test shall be as prescribed in TT-C-490. The test shall be made on the painted bourrelet of each unit of the sample. Visible exposure of underlying phosphate coating or ferrous metal on any unit of the sample as a result of the test shall be classed as a defect.

4.5.4 Corrosion resistance. The method of test shall be as prescribed in American Society for Testing and Materials (ASTM), ASTM B117. The paint shall not be disturbed, however, the rotating band and unpainted surfaces of the sample assemblies may be masked out by a wax coating or other suitable material. The salt spray solution and exposure time shall be as specified in 3.5.3.

4.5.5 Metal parts security. The method of test will be as prescribed by AMCR 715-505, Volume 8, as applicable. The test sample will be inert charged and assembled as specified on Dwg. D7259090, Cartridge, 20mm, M97A2, for projectile metal parts security test. The test cartridges, at ambient temperature, will be fired in bursts of 50 shots in an M24 gun. The gun barrel will be at ambient temperature for the firing of each burst. The cyclic rate of fire will be recorded for informational purposes.

## 5. PACKAGING

5.1 Packaging and packing, level C. Each lot of projectile metal parts shall be packed as specified by Dwg. F7258861 or Dwg. D7259136.

5.2 Marking, level C. Marking or labeling of packed cartons or containers shall comply with applicable requirements of MIL-STD-1169.

## 6. NOTES

6.1 Intended use. These projectile metal parts are intended for use with M21 type cartridge case in the assembly of HEI cartridges for Guns, Automatic, 20mm, M3 and M24.



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6.2 Ordering data. Procurement documents should specify the following:

6.2.1 Title, number and date of this specification.

6.2.2 Place of inspection, if not at place of manufacture.

6.2.3 First article sample requirements. (see 3.3, 4.3 and 6.6)

6.2.4 Packaging, packing and marking instructions. (see 5.1 and 5.2)

6.2.5 Provision for shipment and testing of first article and production lot samples for metal parts security test. (see 4.3 and 4.5.5)

6.2.6 Provision for the supply, maintenance and disposition of Government furnished inspection equipment for acceptance inspection purposes.

6.2.7 Provision for the submission and approval of the manufacturing process. (see 3.2)

6.2.8 Provisions for the inclusion of MIL-STD-1167, Ammunition Data Cards on DD form 1423, Contract Data Requirements List.

6.3 AQL's. The optional use of AQL values for either individual defects or classes of defects, with individual major defect limitation, is intended to minimize inspection agency administrative burden which might result from an exclusive assignment of individual defect AQL's. The option also permits flexibility where sampling inspection for acceptance is integrated into the manufacturing process.

6.4 Intermediate point inspection. The classification of defects identifies the defect characteristics (among other things) for acceptance inspection. It may be necessary to modify the sequence of inspection stations to best suit the manufacturing process. Inspection for defect characteristics which will be hidden or altered by subsequent processing operations (including unrelated operations), should be scheduled to prevent premature acceptance which could be detrimental to the attainment of optimum product quality in the end item.

6.5 Process deviation. A process deviation is defined as a change in the approved basic method of manufacture, or an operational change which alters the metallurgical or physical properties of the item of contract.

6.6 First article sample. Where a first article sample is required by the contract, the contracting officer will notify the testing facility when to proceed with the metal parts security test. Additional assemblies may be required by the test facility. (see 4.3)



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6.7 Changes from previous issues. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

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